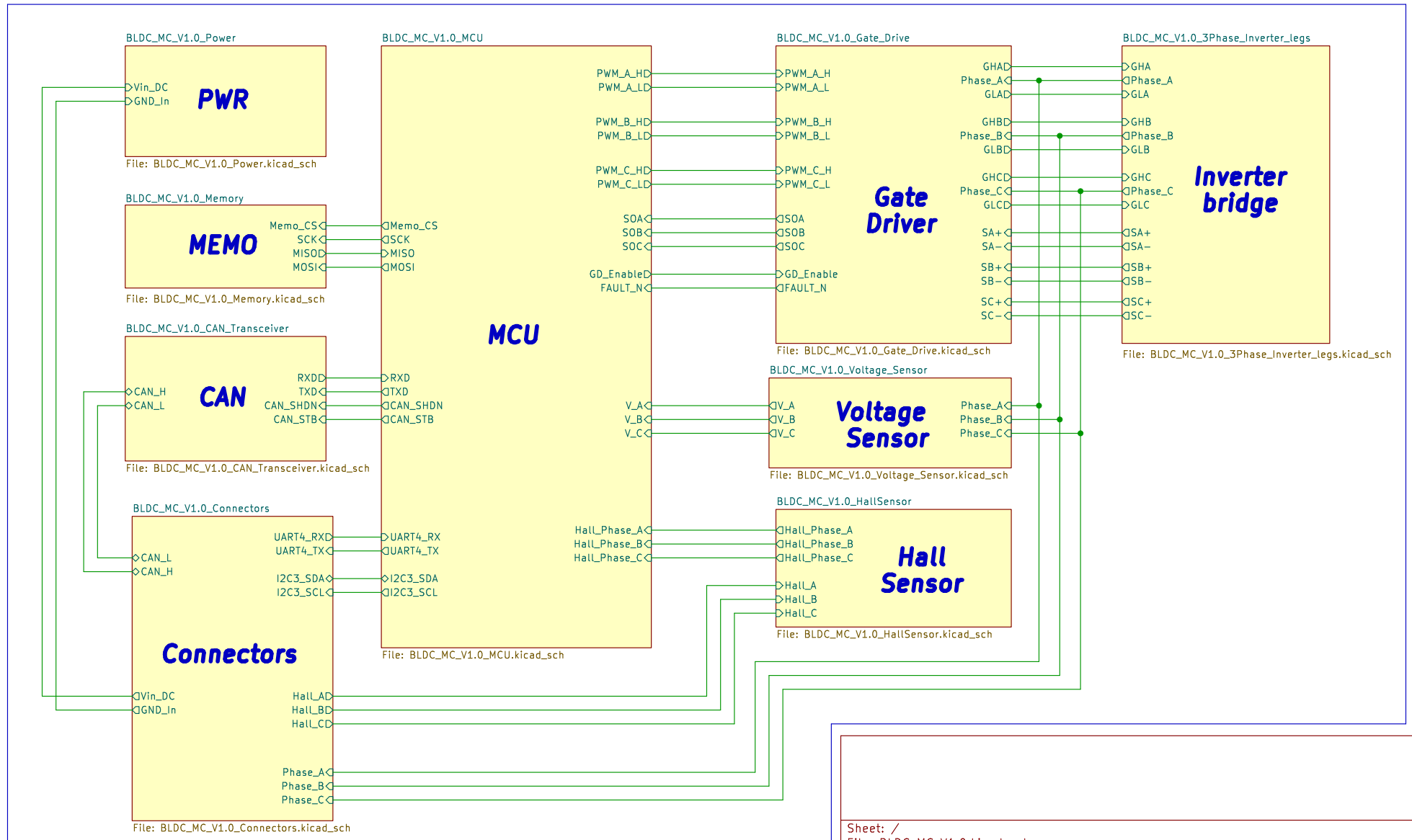


[1] BLDC_MC V1 60V 30A



Sheet: /
File: BLDC_MC_V1.0.kicad_sch

Title: BLDC_MC

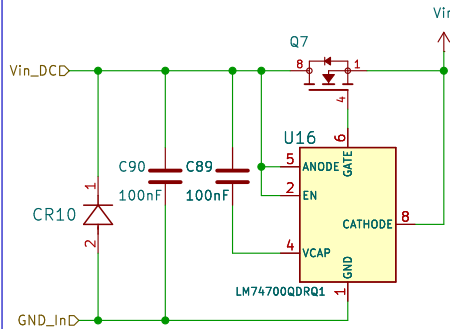
Size: A4 Date: 2026-01-05
KiCad E.D.A. 9.0.7

Rev: V1
Id: 1/10

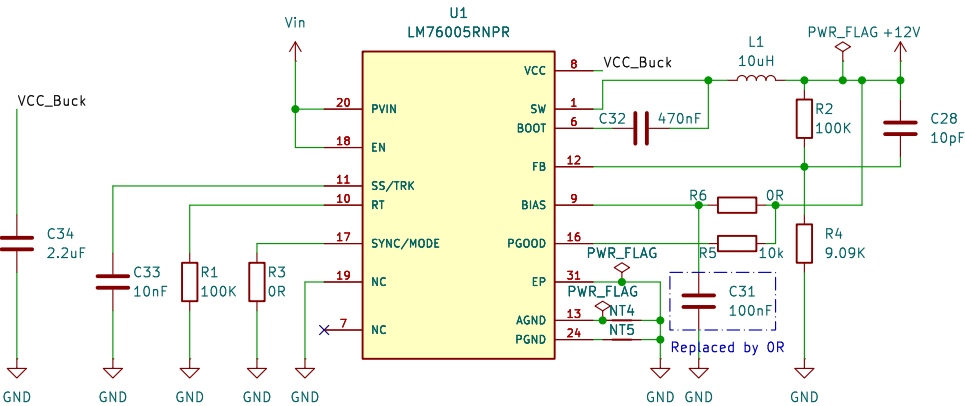
[1] Power Conversion Vin to 12V and 12V to 3.3V

The maximum input vottage of this 60V so we can use for BLDC motors up to 60V

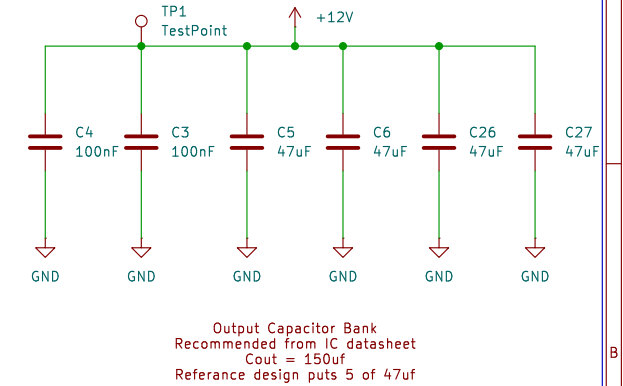
Input PWR Protection



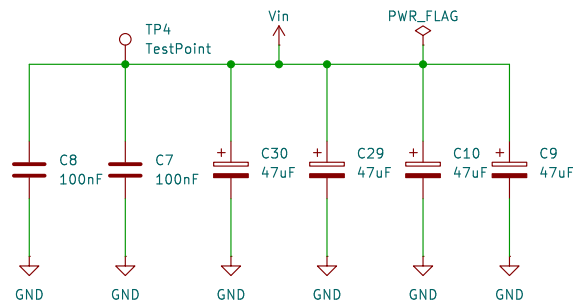
Buck IC Circuit



Output Cap Bank

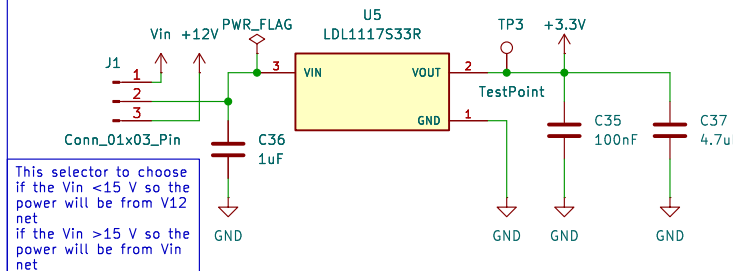


Input Cap Bank

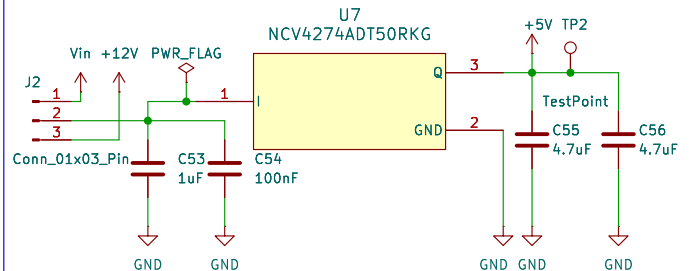


input Capacitor Bank
To make the input power pure DC

LDO Fixed out 3.3V

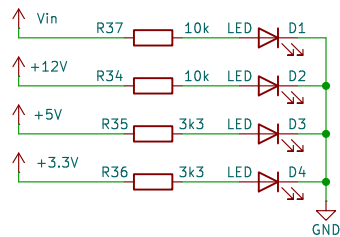


LDO Fixed out 5V



Voltage Level	Usage
12V	Gate drive and MOSFET firing Voltage
5V	for Hall Sensor
3.3V	Controller and aux ICs

Indication Leds



Sheet: /BLDC_MC_V1.0_Power/
File: BLDC_MC_V1.0_Power.kicad_sch

Title: BLDC_MC

Size: A4 Date: 2026-01-05

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Rev: V1

Id: 2/10

6

A



A

B

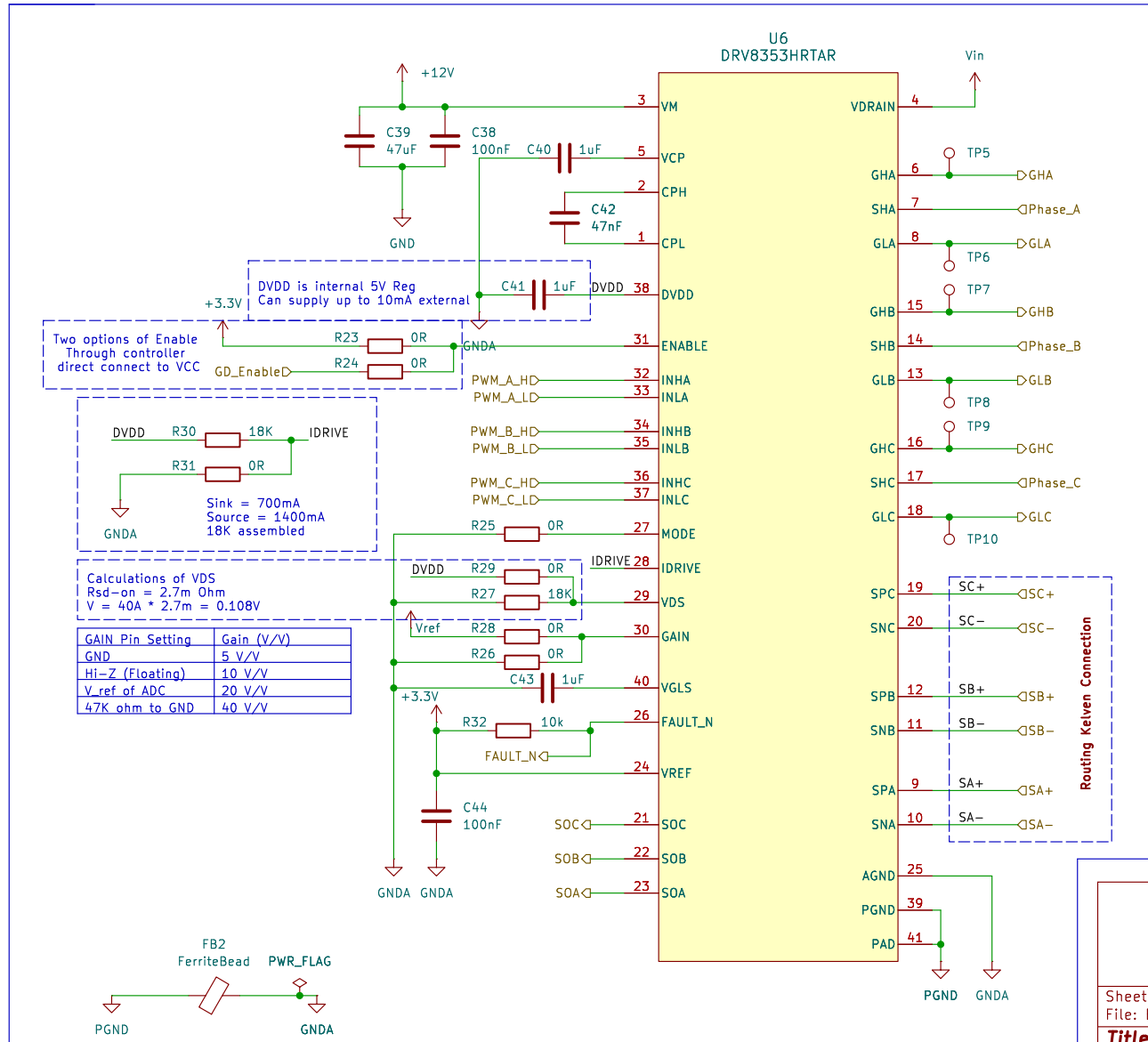
C

C

6

[4] Smart Gate Drive

for the 3 phase inverter with current sensor



Gate Resistors vs. IDRIVENormal Driver:

Uses external gate resistors to limit the current charging the MOSFET gate. If you want to change the switching speed to reduce EMI or heat, you have to desolder and change the resistors.

Smart Driver: Uses an internal, programmable current source called IDRIVE. You set the source/sink current (e.g., 100mA, 1A) via a single pin or SPI. This allows you to tune the switching speed of your inverter without changing any hardware components.

Sheet: /BLDC_MC_V1.0_Gate_Drive/
File: BLDC_MC_V1.0_Gate_Drive.kicad_sch

Title: BLDC_MC

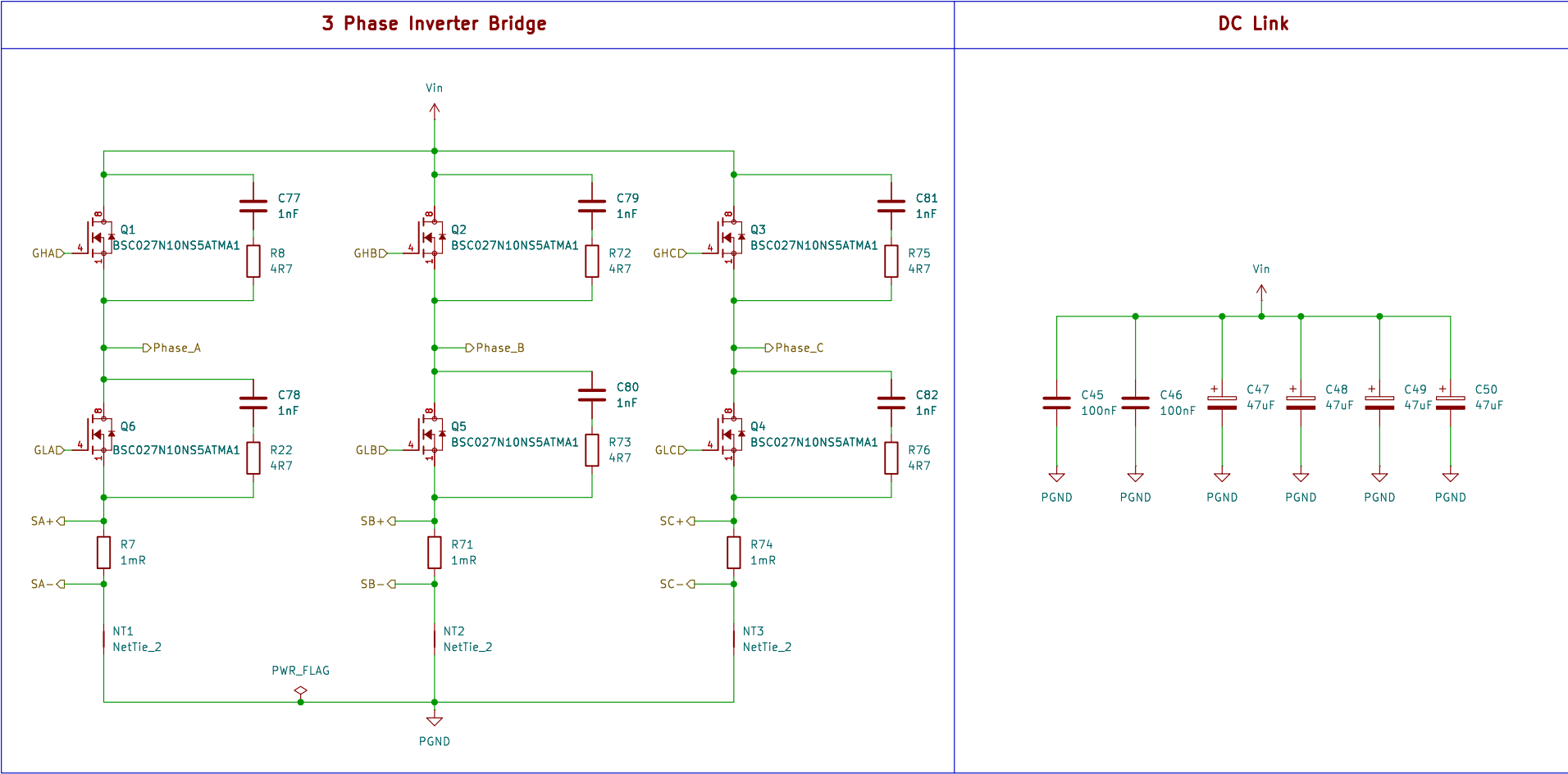
Size: A4
KiCad E.D.A. 9.0.7

Date: 2026-01-12

Rev: V1

Id: 4/10

[5] 3 Phases Inverter Bridge



Sheet: /BLDC_MC_V1.0_3Phase_Inverter_legs/
File: BLDC_MC_V1.0_3Phase_Inverter_legs.kicad_sch

Title: BLDC_MC

Size: A4 Date: 2026-01-07

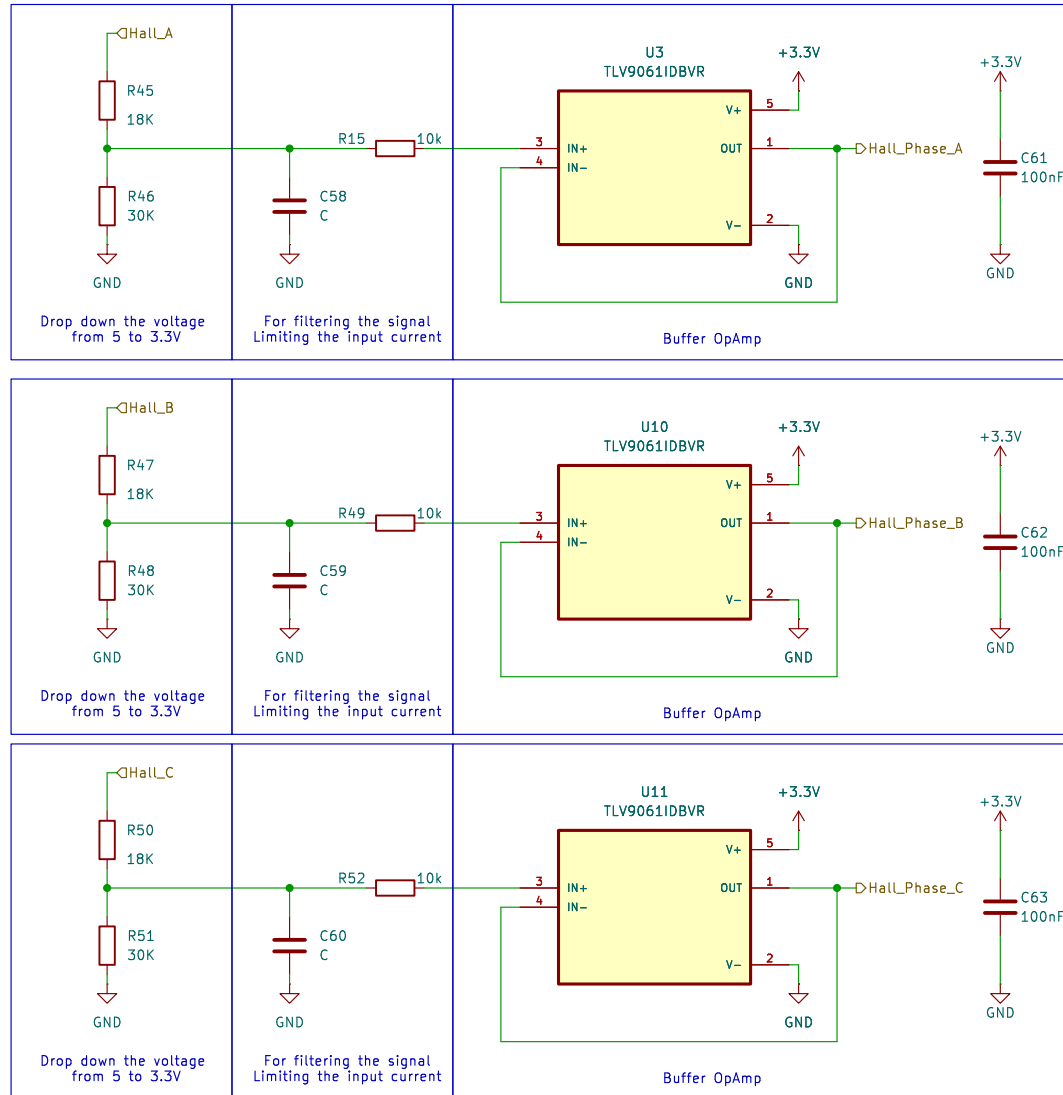
KiCad E.D.A. 9.0.7

Rev: V1

Id: 5/10

[6] Hall Effect Sensor Interface

For the 3 Phases of Hall Sensor



Sheet: /BLDC_MC_V1.0_HallSensor/
File: BLDC_MC_V1.0_HallSensor.kicad_sch

Title: BLDC_MC

Size: A4 Date: 2026-03-05

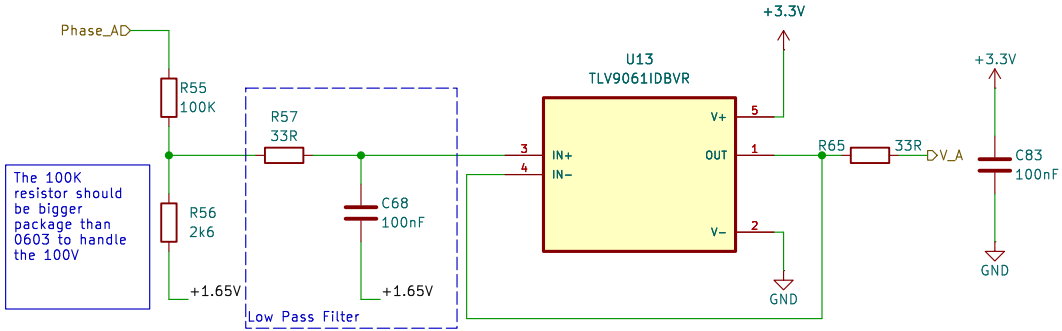
KiCad E.D.A. 9.0.7

Rev: V1

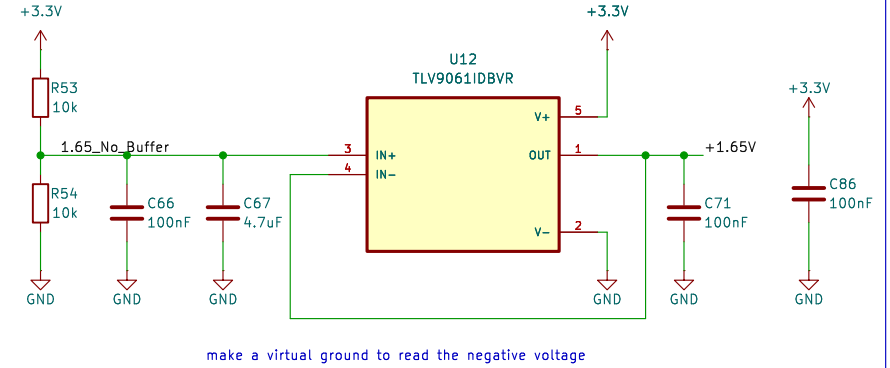
Id: 6/10

[7] 3 Phase Inverter Voltage Sensing

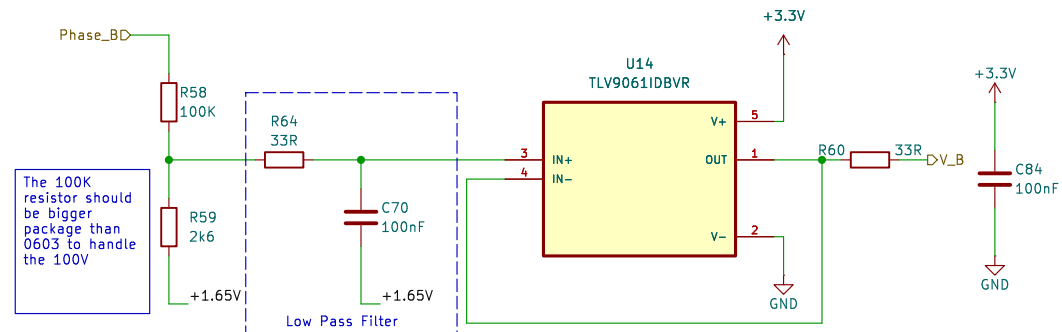
Phase A



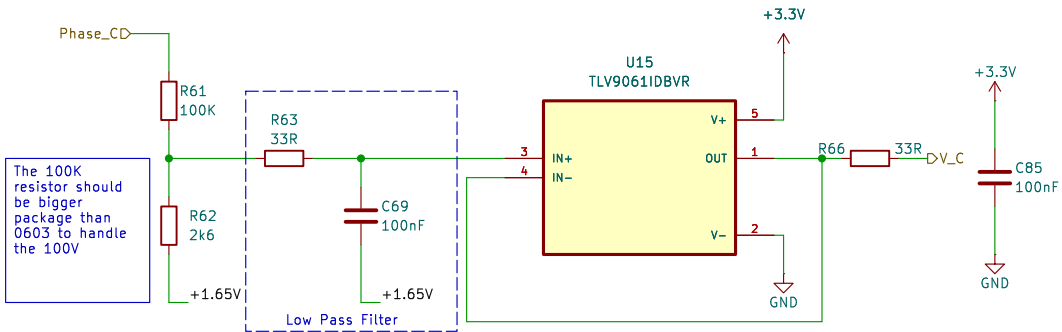
Virtual Ground +1.65V



Phase B



Phase C



Sheet: /BLDC_MC_V1.0_Voltage_Sensor/
File: BLDC_MC_V1.0_Voltage_Sensor.kicad_sch

Title: **BLDC_MC**

Size: A4 Date: 2026-03-11

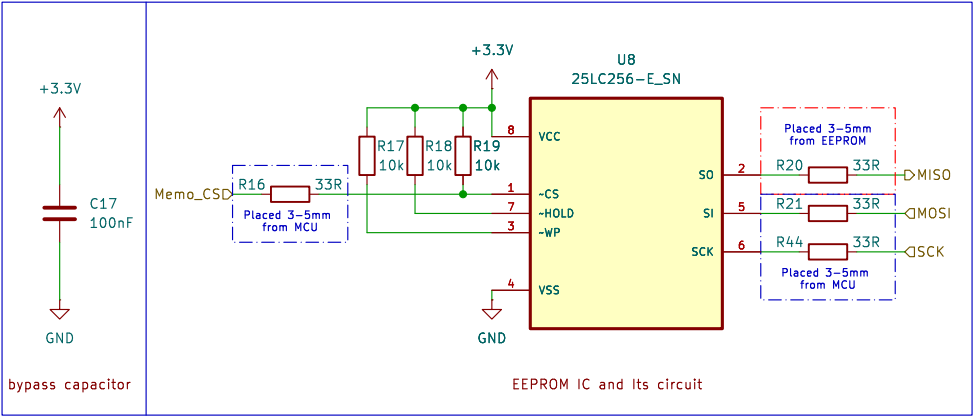
KiCad E.D.A. 9.0.7

Rev: **V1**

Id: 7/10

[8] EEPROM Memory 256Kbit

This memory used to store important data and configration parameters



Sheet: /BLDC_MC_V1.0_Memory/
File: BLDC_MC_V1.0_Memory.kicad_sch

Title: BLDC_MC

Size: A4 Date: 2026-01-07

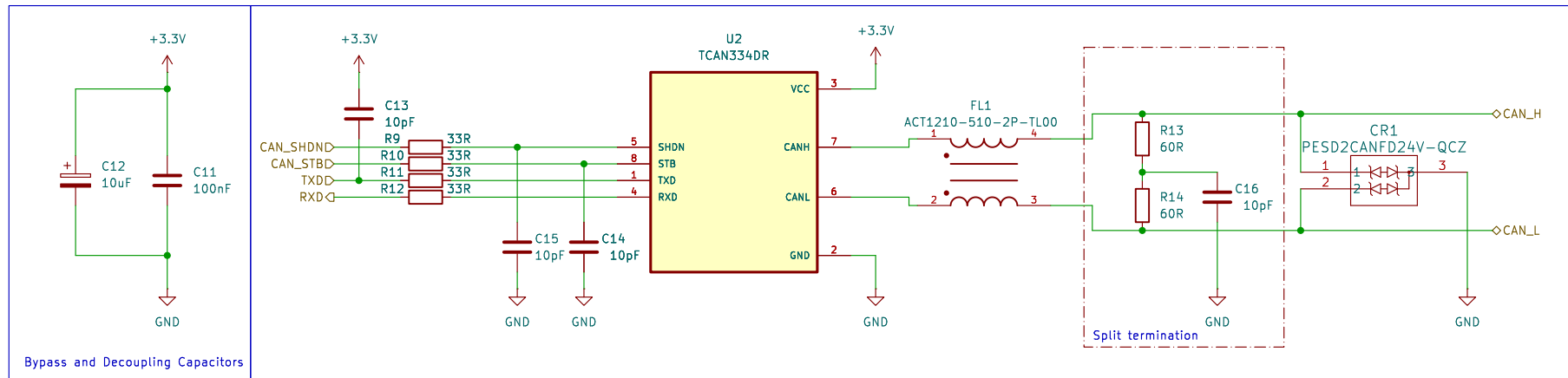
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Rev: V1

Id: 8/10

[9] CAN Transiver Classic CAN 1Mb/S

This IC can operate as CAN-FD for faster data rate



Note

Series Resistors Help manage signal integrity by damping potential high-frequency noise or reflections on the digital traces.

Parallel Capacitors is used to filter noise on the digital I/O lines

Sheet: /BLDC_MC_V1.0_CAN_Transceiver/
File: BLDC_MC_V1.0_CAN_Transceiver.kicad_sch

Title: BLDC_MC

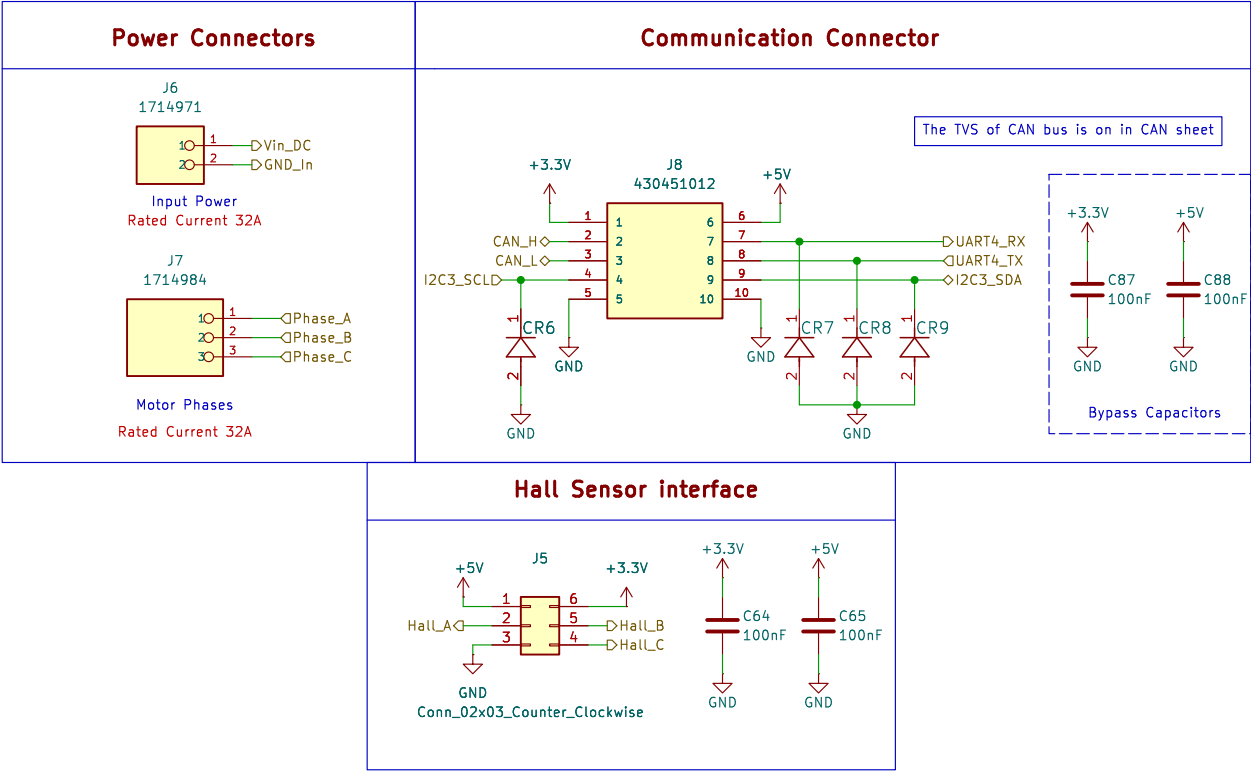
Size: A4 Date: 2026-01-05

KiCad E.D.A. 9.0.7

Rev: V1

Id: 9/10

[10] Board's Connectors Power & Communication



Sheet: /BLDC_MC_V1.0_Connectors/
File: BLDC_MC_V1.0_Connectors.kicad_sch

Title: BLDC_MC

Size: A4 Date: 2026-03-08

KiCad E.D.A. 9.0.7

Rev: V1

Id: 10/10